

$X = \langle \text{Presbyopic, hypermetropia, myopia, normal} \rangle$ onde $f(\cdot) = \text{soft}$

$$V_{NB} = \arg \max_{V_j \in \{\text{more, soft, hard}\}} P(V_j) \prod_i P(a_i | V_j)$$

$$= \arg \max_{V_j \in \{\text{more, soft, hard}\}} P(V_j) P(\text{Presbyopic} | V_j) P(\text{hypermetropia} | V_j) P(\text{myopia} | V_j) P(\text{normal} | V_j)$$

Separando o calculo de cada V_j , temos

$$\underline{V_j = \text{more}} \Rightarrow P(V_j = \text{more}) = \frac{13}{21}$$

$$\left(\frac{13}{21}\right) \cdot P(\text{Presbyopic} | \text{more}) P(\text{hypermetropia} | \text{more}) P(\text{myopia} | \text{more}) P(\text{normal} | \text{more})$$

$$\left(\frac{13}{21}\right) \cdot \left(\frac{4}{13}\right) \left(\frac{6}{13}\right) \left(\frac{7}{13}\right) \left(\frac{2}{13}\right) = \frac{4 \cdot 6 \cdot 7 \cdot 2}{13^3 \cdot 21} = \frac{336}{46137} = 0.7282\%$$

$$\underline{V_j = \text{soft}} \Rightarrow P(V_j = \text{soft}) = \frac{4}{21}$$

$$\left(\frac{4}{21}\right) \cdot \left(\frac{2}{4}\right) \cdot \left(\frac{2}{4}\right) \cdot \left(\frac{4}{4}\right) \cdot \left(\frac{4}{4}\right) = \frac{1}{21} = 0.047619$$

$$\underline{V_j = \text{hard}} \Rightarrow P(V_j = \text{hard}) = \frac{4}{21}$$

$$\left(\frac{4}{21}\right) \cdot \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) \left(\frac{0}{4}\right) \left(\frac{4}{4}\right) = 0$$

$$v_{NB} = \arg \max_{v_j \in V} (0.0072, 0.0476, 0) = \text{soft}$$

2ª instância

$$x = \langle \text{presbyopic}, \text{hypermetropic}, \text{yes}, \text{reduced} \rangle, \text{ onde } y = \text{more}$$

$$v_{NB} = \arg \max_{v_j \in \{\text{more}, \text{soft}, \text{hard}\}} P(v_j) \prod_i P(a_i | v_j)$$

$$= \arg \max_{v_j \in \{\text{more}, \text{soft}, \text{hard}\}} P(v_j) P(\text{presbyopic} | v_j) P(\text{hypermetropic} | v_j) P(\text{yes} | v_j) P(\text{reduced} | v_j)$$

Para $v_j = \text{more} \Rightarrow P(v_j = \text{more}) = \frac{13}{21}$

$$= \left(\frac{13}{21} \right) \left(\frac{4}{13} \right) \left(\frac{6}{13} \right) \left(\frac{6}{13} \right) \left(\frac{11}{13} \right) = \frac{46 \cdot 6}{21 \cdot 13^3} = \frac{144}{46137} = 0.0031211$$

Para $v_j = \text{soft} \quad P(\text{soft}) = \frac{4}{21}$

$$= \left(\frac{4}{21} \right) \left(\frac{2}{4} \right) \left(\frac{2}{4} \right) \left(\frac{0}{4} \right) \left(\frac{0}{4} \right) = 0$$

Para $v_j = \text{hard} \quad P(v_j = \text{hard}) = \frac{4}{21}$

$$= \left(\frac{4}{21} \right) \left(\frac{1}{4} \right) \left(\frac{1}{4} \right) \left(\frac{4}{4} \right) \left(\frac{0}{4} \right) = 0$$

$$v_{NB} = \arg \max_{v_j \in V} (0.003121, 0, 0) = \text{mome}$$

3ª instância

$X = \langle \text{presbyopic, hypermetropic, yes, normal} \rangle$, com $Y = \text{mome}$

$$v_{NB} = \arg \max_{v_j \in V} P(v_j) \prod_i P(a_i | v_j)$$

Para $v_j = \text{mome}$ $P(\text{mome}) = \frac{13}{21}$

$$\left(\frac{13}{21}\right) \left(\frac{4}{13}\right) \left(\frac{6}{13}\right) \left(\frac{4}{13}\right) \left(\frac{2}{13}\right) = \frac{4.64}{21 \cdot 13^3} = 0.7282\%$$

Para $v_j = \text{soft}$

$$\left(\frac{4}{21}\right) \left(\frac{2}{4}\right) \left(\frac{2}{4}\right) \left(\frac{0}{4}\right) \left(\frac{0}{4}\right) = 0$$

Para $v_j = \text{hard termos}$ $P(\text{yes} | \text{hard}) = 0$, o que torna $P(\text{hard}) \prod_i P(a_i | \text{hard}) = 0$.

Logo, $v_{NB} = \text{mome}$.